

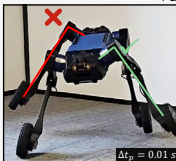
(a) Two-Leg Stance Jumping

VBAC

Proposed



Position Limit



Unstable / Fail



$\Delta t_p = 0.05 \text{ s}$



$\Delta t_p = 0.01 \text{ s}$



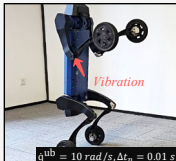
$\Delta t_p = 0.05 \text{ s}$

Stable

(b) Bipedal Standing



Velocity Limit



$\dot{q}^{ub} = 10 \text{ rad/s}, \Delta t_p = 0.01 \text{ s}$

Success / Fail



$\dot{q}^{ub} = 8 \text{ rad/s}, \Delta t_p = 0.01 \text{ s}$



$\dot{q}^{ub} = 10 \text{ rad/s}, \Delta t_p = 0.01 \text{ s}$



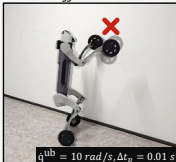
$\dot{q}^{ub} = 8 \text{ rad/s}, \Delta t_p = 0.01 \text{ s}$

Success / Standing but Uncontrollable

(c) Bipedal Standing with Upper Limb End-Effector Motion



Position + Velocity Limit

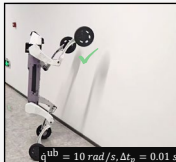


$\dot{q}^{ub} = 10 \text{ rad/s}, \Delta t_p = 0.01 \text{ s}$

Oscillation / Fail



$\dot{q}^{ub} = 10 \text{ rad/s}, \Delta t_p = 0.05 \text{ s}$



$\dot{q}^{ub} = 10 \text{ rad/s}, \Delta t_p = 0.01 \text{ s}$



$\dot{q}^{ub} = 10 \text{ rad/s}, \Delta t_p = 0.05 \text{ s}$

Stable / Success